Faculdade de Design,
Tecnologia e Comunicação
Universidade Europeia

Resources

Automatic Applied Learning

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Agenda

- IADE resources
- Python resources

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IADE Resources

- **Github Education** – https://github.com/education

- **Office 365** – https://portal.office.com

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Python Resources



- https://app.edublocks.org



- https://www.w3schools.com/python/python_intro.asp

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Python Resources

VS Code

Homepage
https://code.visualstudio.com

Documentation
https://code.visualstudio.com/docs

python™

Homepage
https://www.python.org

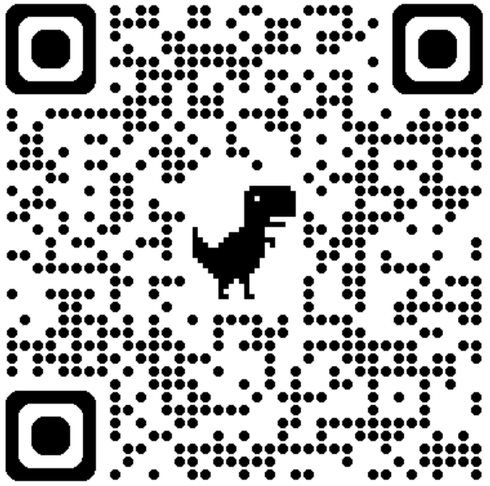
Documentation
https://www.python.org/doc

or

ANACONDA

Homepage
https://www.anaconda.com

Documentation
https://www.anaconda.com/docs



PyTorch

Homepage
https://pytorch.org/projects/pytorch

Documentation
https://docs.pytorch.org/docs/stable

scikit learn

Homepage
https://scikit-learn.org/

Documentation
https://scikit-learn.org/stable/user_guide.html

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Python Resources

Python Virtual Environment

- https://www.w3schools.com/python/python_virtualenv.asp

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Python Resources

- **Creating a Virtual Environment**
  - Choose a directory and then execute the follow command:

    python -m venv ml



  - To **change directory** and **list content**, execute the follow command:

    cd ml
    ls

```
(base) PS D:\programming\IADE\IADE-MCCAI-AML-26> cd ml
(base) PS D:\programming\IADE\IADE-MCCAI-AML-26\ml> ls


    Diretório: D:\programming\IADE\IADE-MCCAI-AML-26\ml


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        30/09/2026     17:14                Include
d-----        30/09/2026     17:14                Lib
d-----        30/09/2026     17:15                Scripts
-a----        30/09/2026     17:14             71 .gitignore
-a----        30/09/2026     17:14            235 pyvenv.cfg
```

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Python Resources

- **To activate** the virtual environment, inside directory ml:

  . \Scripts\activate

```
(base) PS D:\programming\IADE\IADE-MCCAI-AML-26\ml> .\Scripts\activate
(ml) (base) PS D:\programming\IADE\IADE-MCCAI-AML-26\ml>
```

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Python Resources

- To update pip (installer)

  python -m pip install --upgrade pip

```
(ml) (base) PS D:\programming\IADE\IADE-MCCAI-AML-26\ml> python -m pip install --upgrade pip
Requirement already satisfied: pip in .\Lib\site-packages (26.1.2)
Collecting pip
  Downloading pip-26.2.1-py3-none-any.whl.metadata (4.6 kB)
Downloading pip-26.2.1-py3-none-any.whl (1.8 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 1.8/1.8 MB 1.4 MB/s  0:00:01
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 26.1.2
    Uninstalling pip-26.1.2:
      Successfully uninstalled pip-26.1.2
Successfully installed pip-26.2.1
```

- To install packages

  pip install package_1_name package_2_name …

  Example:

  pip install torch scikit-learn matplotlib

```
(ml) (base) PS D:\programming\IADE\IADE-MCCAI-AML-26\ml> pip install torch scikit-learn matplotlib
Collecting torch
  Downloading torch-2.14.0-cp314-cp314-win_amd64.whl.metadata (38 kB)
Collecting scikit-learn
  Downloading scikit_learn-1.9.1-cp314-cp314-win_amd64.whl.metadata (9.3 kB)
Collecting matplotlib
  Downloading matplotlib-3.11.2-cp314-cp314-win_amd64.whl.metadata (80 kB)
Collecting filelock (from torch)
  Downloading filelock-4.0.7-py3-none-any.whl.metadata (2.0 kB)
Collecting typing-extensions>=4.10.0 (from torch)
  Downloading typing_extensions-4.16.0-py3-none-any.whl.metadata (3.3 kB)
Collecting setuptools>=77.0.3 (from torch)
  Downloading setuptools-84.0.0-py3-none-any.whl.metadata (6.6 kB)
Collecting sympy>=1.13.3 (from torch)
  Downloading sympy-1.14.0-py3-none-any.whl.metadata (12 kB)
Collecting networkx>=2.5.1 (from torch)
  Downloading networkx-3.7-py3-none-any.whl.metadata (6.7 kB)
Collecting jinja2 (from torch)
  Downloading jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting fsspec>=0.8.5 (from torch)
  Downloading fsspec-2026.9.0-py3-none-any.whl.metadata (10 kB)
Collecting numpy>=1.24.1 (from scikit-learn)
  Downloading numpy-2.5.3-cp314-cp314-win_amd64.whl.metadata (6.6 kB)
Collecting scipy>=1.10.0 (from scikit-learn)
  Downloading scipy-1.18.1-cp314-cp314-win_amd64.whl.metadata (61 kB)
Collecting joblib>=1.4.0 (from scikit-learn)
  Downloading joblib-1.6.0-py3-none-any.whl.metadata (6.5 kB)
Collecting narwhals>=2.0.1 (from scikit-learn)
  Downloading narwhals-2.26.0-py3-none-any.whl.metadata (15 kB)
Collecting threadpoolctl>=3.5.0 (from scikit-learn)
  Downloading threadpoolctl-3.7.0-py3-none-any.whl.metadata (24 kB)
Collecting contourpy>=1.0.1 (from matplotlib)
  Downloading contourpy-1.4.0-cp314-cp314-win_amd64.whl.metadata (3.8 kB)
Collecting cycler>=0.10 (from matplotlib)
  Downloading cycler-0.12.1-py3-none-any.whl.metadata (3.8 kB)
Collecting fonttools>=4.28.2 (from matplotlib)
  Downloading fonttools-4.66.1-cp314-cp314-win_amd64.whl.metadata (132 kB)
Collecting kiwisolver>=1.3.1 (from matplotlib)
  Downloading kiwisolver-1.5.1-cp314-cp314-win_amd64.whl.metadata (5.2 kB)
Collecting packaging>=20.0 (from matplotlib)
  Downloading packaging-26.3-py3-none-any.whl.metadata (3.5 kB)
Collecting pillow>=9 (from matplotlib)
  Downloading pillow-12.3.0-cp314-cp314-win_amd64.whl.metadata (9.3 kB)
Collecting pyparsing>=3 (from matplotlib)
  Downloading pyparsing-3.3.3-py3-none-any.whl.metadata (5.9 kB)
Collecting python-dateutil>=2.7 (from matplotlib)
  Downloading python_dateutil-2.9.0.post0-py2.py3-none-any.whl.metadata (8.4 kB)
Collecting cloudpickle>=3.0 (from joblib>=1.4.0->scikit-learn)
  Downloading cloudpickle-3.1.2-py3-none-any.whl.metadata (7.1 kB)
Collecting six>=1.5 (from python-dateutil>=2.7->matplotlib)
  Downloading six-1.17.0-py2.py3-none-any.whl.metadata (1.7 kB)
Collecting mpmath<1.4,>=1.1.0 (from sympy>=1.13.3->torch)
  Downloading mpmath-1.3.0-py3-none-any.whl.metadata (8.6 kB)
Collecting MarkupSafe>=2.0 (from jinja2->torch)
  Downloading markupsafe-3.0.3-cp314-cp314-win_amd64.whl.metadata (2.8 kB)
Downloading torch-2.14.0-cp314-cp314-win_amd64.whl (124.1 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 85.7/124.1 MB 2.8 MB/s eta 0:00:14
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